Generative Modelling with Stochastic Differential Equations

Exercises, week 4

Convergence of diffusion models

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Exercises, chapters and parts marked ($\star$) are on material that is not covered in the lectures and is left for self-study. It is examinable like the rest.

6 Convergence of diffusion models

Setting. Throughout, q is a probability distribution on $\mathbb{R}^n$, $m_2^2 := \mathbb{E}_{X\sim q}|X|^2$ (possibly infinite), $\gamma_n = \mathcal{N}(0, \mathbb{I}_n)$, and $\bar{X}_\tau$ is the forward Ornstein–Uhlenbeck (OU) process

$$\mathrm{d}\bar{X}_\tau = -\bar{X}_\tau \,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}B_\tau, \qquad \bar{X}_0 \sim q,$$

with marginals $q_\tau = \mathrm{Law}(\bar{X}_\tau)$, scores $s_\tau = \nabla \log q_\tau$ (smooth for $\tau > 0$) and $\sigma_\tau^2 = 1 - e^{-2\tau}$. You may use the following facts.

- *Explicit solution.* For each fixed $\tau > 0$, $\bar{X}_\tau = e^{-\tau}\bar{X}_0 + \sigma_\tau Z_\tau$, where $Z_\tau := (\bar{X}_\tau - e^{-\tau}\bar{X}_0)/\sigma_\tau \sim \gamma_n$ is independent of $\bar{X}_0$. The variable Z_τ depends on τ, and as a process the solution is the stochastic convolution $e^{-\tau}\bar{X}_0 + \sqrt{2}\int_0^\tau e^{-(\tau-r)}\,\mathrm{d}B_r$.
- *Markov form.* For $0 \le \tau' \le \tau$, $\bar{X}_\tau = e^{-(\tau-\tau')}\bar{X}_{\tau'} + \sigma_{\tau-\tau'}Z'$ with $Z' \sim \gamma_n$ independent of $(\bar{X}_r)_{r\le\tau'}$.
- *Posterior formulas.* With $\Sigma_\tau(x) = \mathrm{Cov}(\bar{X}_0 \mid \bar{X}_\tau = x)$ and $a(\tau) = \mathbb{E}\,\mathrm{tr}\,\Sigma_\tau(\bar{X}_\tau)$,

  $$s_\tau(x) = -\sigma_\tau^{-1}\mathbb{E}[Z \mid \bar{X}_\tau = x], \qquad \nabla^2 \log q_\tau(x) = -\sigma_\tau^{-2}\mathbb{I}_n + e^{-2\tau}\sigma_\tau^{-4}\Sigma_\tau(x),$$

  and $\mathbb{E}|s_\tau(\bar{X}_\tau)|^2 = n\sigma_\tau^{-2} - e^{-2\tau}\sigma_\tau^{-4}a(\tau)$.

The sampler. Fix $T > 0$, a reverse-time grid $0 = t_0 < \cdots < t_K \le T$ with steps $h_k = t_{k+1} - t_k$, and a learned score $\hat{s}_\tau \approx s_\tau$. The exponential-integrator sampler is the process $\hat{Y}$ defined on each $[t_k, t_{k+1}]$ by

$$\mathrm{d}\hat{Y}_t = \big[\hat{Y}_t + 2\hat{s}_{T-t_k}(\hat{Y}_{t_k})\big]\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \qquad \hat{Y}_0 \sim \gamma_n, \tag{6.1}$$

with B a Brownian motion independent of $\hat{Y}_0$. It approximates the reverse process $\overleftarrow{X}_t = \bar{X}_{T-t}$, whose drift is $\overleftarrow{X}_t + 2s_{T-t}(\overleftarrow{X}_t)$. Two results on its accuracy are referred to below. The *Lipschitz theorem* says that if every s_τ is L-Lipschitz with $L \ge 1$, $t_K = T \ge 1$ and $h = \max_k h_k$, then $\mathrm{KL}(q \,\|\, \mathrm{Law}(\hat{Y}_T)) \le (\frac{1}{2}m_2^2 + ne^{-2T})e^{-2T} + 3L^2nhT + 2\varepsilon_{\text{score}}^2$, where $\varepsilon_{\text{score}}^2$ is the integrated score error $\sum_k h_k \mathbb{E}_{q_{T-t_k}}|\hat{s}_{T-t_k} - s_{T-t_k}|^2$. The *stochastic-localisation theorem* needs no smoothness, assumes $m_2^2 \le n$ in its simplified form, stops at $t_K = T - \delta$, and uses a grid with $h_k \le \kappa \min\{1, T - t_{k+1}\}$.

Exercise 6.1 (Exponential-integrator update)**.** Let $h > 0$ and let $c, y \in \mathbb{R}^n$ be fixed.

(a) Show that the solution of $\mathrm{d}Y_t = (Y_t + c)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, $Y_0 = y$, satisfies

$$Y_h = e^h y + (e^h - 1)c + \sqrt{2}\int_0^h e^{h-r}\,\mathrm{d}B_r \sim \mathcal{N}(e^h y + (e^h - 1)c,\ (e^{2h} - 1)\mathbb{I}_n).$$

Hint: apply Itô's formula to $e^{-t}Y_t$.

(b) Deduce that the scheme (6.1) satisfies

$$\hat{Y}_{t_{k+1}} = e^h \hat{Y}_{t_k} + 2(e^h - 1)\,\hat{s}_{T-t_k}(\hat{Y}_{t_k}) + \sqrt{e^{2h} - 1}\,\xi_k,$$

where $\xi_0, \ldots, \xi_{K-1}$ are i.i.d. γ_n and independent of $\hat{Y}_0$.

(c) Show that the Euler–Maruyama step $\hat{Y}_{t_k} + h\big(\hat{Y}_{t_k} + 2\hat{s}_{T-t_k}(\hat{Y}_{t_k})\big) + \sqrt{2h}\,\xi_k$ agrees with the update in (b) up to $O(h^2)$ in the drift and up to a relative $O(h)$ in the noise standard deviation.

Exercise 6.2 (Movement bound)**.** (a) Show that if $0 \le \tau' \le \tau$ and $u = \tau - \tau' \le 1$, then $\mathbb{E}|\bar{X}_\tau - \bar{X}_{\tau'}|^2 \le m_2^2 u^2 + 3nu$. *Hint:* use the Markov form of the OU process and the moment bound $\mathbb{E}_{q_\tau}|X|^2 \le n \vee m_2^2$ of Exercise 6.3(a).

(b) Where is $u \le 1$ used? What bound holds for $u > 1$?

(c) Let $q = \gamma_n$. Compute $\mathbb{E}|\bar{X}_\tau - \bar{X}_{\tau'}|^2$ exactly and show that it is $\sim 2nu$ as $u \to 0$. Deduce that, if s_τ is L-Lipschitz, then $\mathbb{E}|s_\tau(\bar{X}_\tau) - s_\tau(\bar{X}_{\tau'})|^2 \lesssim L^2 nu$ for small u, and explain why the discretisation error in the Lipschitz theorem recalled above is of order $L^2 nhT$ in KL, that is $L\sqrt{nhT}$ in total variation, rather than $L^2 h^2 T$.

Exercise 6.3 (Moment and score bounds)**.** (a) Show that $\mathbb{E}_{q_\tau}|X|^2 = e^{-2\tau} m_2^2 + \sigma_\tau^2 n \le n \vee m_2^2$.

(b) Let p be a positive C^1 probability density on $\mathbb{R}^n$ such that $G := \nabla \log p$ is L-Lipschitz. Show that for every nonnegative $\psi \in C_c^1(\mathbb{R}^n)$,

$$\int \langle G, \nabla\psi \rangle \,\mathrm{d}x \le nL \int \psi \,\mathrm{d}x.$$

Hint: write $\partial_i \psi$ as a limit of difference quotients and move the shift onto G.

(c) Deduce that $\mathbb{E}_p|\nabla \log p|^2 \le nL$, and hence that $\mathbb{E}_{q_\tau}|s_\tau|^2 \le nL$ whenever s_τ is L-Lipschitz. *Hint:* take $\psi = \varphi_R\, p$ with a cutoff φ_R.

(d) Explain why (b)–(c) are a rigorous version of the formal computation $\mathbb{E}_p|\nabla \log p|^2 = -\mathbb{E}_p[\Delta \log p] \le nL$.

(e) Show that equality holds in (c) for $p = \mathcal{N}(0, L^{-1}\mathbb{I}_n)$.

Exercise 6.4 (Two symmetric mixtures)**.** Let $n \ge 2$ and $\mathbf{1}_n = (1, \dots, 1)^\top$.

(a) Let $q = \frac{1}{2}\mathcal{N}(\mathbf{1}_n, \mathbb{I}_n) + \frac{1}{2}\mathcal{N}(-\mathbf{1}_n, \mathbb{I}_n)$. Show that $q_\tau = \frac{1}{2}\mathcal{N}(e^{-\tau}\mathbf{1}_n, \mathbb{I}_n) + \frac{1}{2}\mathcal{N}(-e^{-\tau}\mathbf{1}_n, \mathbb{I}_n)$, compute s_τ and $\nabla^2 \log q_\tau$, and show that $\sup_{\tau \ge 0} \sup_x \|\nabla^2 \log q_\tau(x)\|_{\mathrm{op}} = n - 1$ when $n \ge 2$. Compute $\mathrm{Cov}(q)$ and $\mathbb{E}|\bar{X}_0|^2$.

(b) Let $q = \frac{1}{2}\delta_{\mathbf{1}_n} + \frac{1}{2}\delta_{-\mathbf{1}_n}$. Compute q_τ, s_τ, $\nabla^2 \log q_\tau$, the posterior mean m_τ and the posterior covariance Σ_τ, and check them against the posterior formulas $s_\tau(x) = -\sigma_\tau^{-2} x + e^{-\tau}\sigma_\tau^{-2} m_\tau(x)$ and $\nabla^2 \log q_\tau(x) = -\sigma_\tau^{-2}\mathbb{I}_n + e^{-2\tau}\sigma_\tau^{-4}\Sigma_\tau(x)$. Show that the Lipschitz constant L_τ of s_τ satisfies $n e^{-2\tau}\sigma_\tau^{-4} - \sigma_\tau^{-2} \le L_\tau \le n e^{-2\tau}\sigma_\tau^{-4} + \sigma_\tau^{-2}$.

(c) An analysis based on a global Lipschitz constant L of all the scores s_τ, $\tau \ge 0$ (Chen et al., 2022) needs $K \asymp L^2 n/\varepsilon^2$ steps (up to logarithms) for accuracy ε, whereas the stochastic-localization analysis (Benton et al., 2024) needs $K \asymp n(T + \log(1/\delta))^2/\varepsilon^2$ steps to approximate the early-stopped q_δ, assuming only $m_2^2 \lesssim n$, with $T \asymp \log(n/\varepsilon^2)$. Compare the two for these examples.

Exercise 6.5 (Extreme cases of the score bounds)**.** (a) Show that $\mathbb{E}|s_\tau(\bar{X}_\tau)|^2$ is at most $n\sigma_\tau^{-2}$, and that equality holds if and only if q is a Dirac mass.

(b) Let $q = \gamma_n$. Compute Σ_τ, $a(\tau)$ and

$$b(\tau) := \mathbb{E}\,\mathrm{tr}(\Sigma_\tau(\bar{X}_\tau)^2),$$

and check the identity $a'(\tau) = 2e^{-2\tau}\sigma_\tau^{-4}b(\tau)$. Show that the bounds $a(\tau) \le n(e^{2\tau}-1)$ and $a(\tau) \le \mathrm{tr}\,\mathrm{Cov}(q)$ are sharp up to constants, for small and for large τ respectively.

(c) Let $q = \mathcal{N}(0, v\mathbb{I}_n)$ with $v > 0$. Show that s_τ is L_τ-Lipschitz with $L_\tau = 1/(e^{-2\tau}v + \sigma_\tau^2)$, and that $\mathbb{E}|s_\tau(\bar{X}_\tau)|^2 = nL_\tau$.

Exercise 6.6 (Early stopping)**.** (a) Show that $\mathcal{W}_2(q_\delta, q)^2 \le (1-e^{-\delta})^2 m_2^2 + n\sigma_\delta^2$, and that the rate $\sqrt{n\delta}$ cannot be improved in general, by computing $\mathcal{W}_2(q_\delta, q)$ for $q = \delta_0$.

(b) Assume $m_2^2 = n$. Given $\varepsilon' > 0$, how small must δ be to guarantee $\mathcal{W}_2(q_\delta, q) \le \varepsilon'$? What does this cost in the step count $K \asymp n(T + \log(1/\delta))^2/\varepsilon^2$, $T = \frac{1}{2}\log(n/\varepsilon^2)$, needed by the stochastic-localisation theorem to approximate q_δ within ε^2 in KL?

Exercise 6.7 (The step schedule)**.** Let $T > 1$, $\delta \in (0,1)$, and let K be even with $K \ge 2\log(1/\delta)$.

(a) Define $t_k = 2k(T-1)/K$ for $0 \le k \le K/2$, and $T - t_{K/2+j} = \delta^{2j/K}$ for $0 \le j \le K/2$. Show that this is a well-defined grid with $t_{K/2} = T-1$ and $t_K = T-\delta$, and that it satisfies the step-size condition

$$h_k \le \kappa \min\{1, T - t_{k+1}\}, \qquad k = 0, \dots, K-1,$$

with $\kappa = 4(T + \log(1/\delta))/K$.

(b) Conversely, show that any grid from 0 to $T-\delta$ that contains $T-1$ and satisfies the step-size condition has $\kappa K \ge T - 1 + \log(1/\delta)$. So the schedule in (a) is optimal up to constants.

Exercise 6.8 (De Bruijn identity and entropy decay)**.** Assume that q has a smooth positive density and that all densities below are smooth enough and decay fast enough to justify differentiation under the integral sign and integration by parts. Let $f_\tau = q_\tau/\gamma_n$ and let $\mathrm{I}(q_\tau\|\gamma_n) = \int|\nabla \log f_\tau|^2\,\mathrm{d}q_\tau$ be the relative Fisher information.

(a) Show that the Fokker–Planck equation of the OU process, $\partial_\tau q_\tau = \nabla\cdot(xq_\tau) + \Delta q_\tau$, can be written $\partial_\tau q_\tau = \nabla\cdot(q_\tau \nabla \log f_\tau)$.

(b) Prove the de Bruijn identity $\frac{\mathrm{d}}{\mathrm{d}\tau}\mathrm{KL}(q_\tau\|\gamma_n) = -\mathrm{I}(q_\tau\|\gamma_n)$.

(c) Using the Gaussian log-Sobolev inequality $\mathrm{KL}(\mu\|\gamma_n) \le \frac{1}{2}\mathrm{I}(\mu\|\gamma_n)$ (valid for every μ with smooth positive density), deduce $\mathrm{KL}(q_T\|\gamma_n) \le e^{-2T}\mathrm{KL}(q\|\gamma_n)$, and hence the initialisation bound $\mathrm{TV}(\gamma_n, q_T) \le \sqrt{\mathrm{KL}(q\|\gamma_n)/2}\,e^{-T}$ (Pinsker's inequality: $\mathrm{TV}(\mu,\nu) \le \sqrt{\mathrm{KL}(\mu\|\nu)/2}$). Compare with the bound $\mathrm{KL}(q_T\|\gamma_n) \le \frac{1}{2}(e^{-2T}(m_2^2 - n) - n\log\sigma_T^2)$, valid whenever $m_2 < \infty$. Which one applies when $\mathrm{KL}(q\|\gamma_n) = \infty$?

(d) Check (b) and (c) for $q = \mathcal{N}(m, \mathbb{I}_n)$ and show that the rate e^{-2T} in (c) is attained.

Exercise 6.9 (A parametric contrast)**.** Let $q = \mathcal{N}(\mu, \mathbb{I}_n)$ with unknown $\mu \in \mathbb{R}^n$, and $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} q$. For an estimator $\hat{s}_\tau$ built from the sample, let $R_\tau(\hat{s}_\tau) := \mathbb{E}_{X\sim q_\tau}|\hat{s}_\tau(X) - s_\tau(X)|^2$, with X independent of the sample.

(a) Compute q_τ and s_τ.

(b) For the plug-in estimator $\hat{s}_\tau(x) = -(x - e^{-\tau}\hat{\mu})$ with $\hat{\mu} = \frac{1}{N}\sum_i X_i$, show $\mathbb{E}R_\tau(\hat{s}_\tau) = e^{-2\tau}n/N$.

(c) Show that every estimator $\hat{s}_\tau$ (any measurable function of the sample and of the evaluation point) satisfies $\sup_\mu \mathbb{E}R_\tau(\hat{s}_\tau) \geq e^{-2\tau}n/(N+1)$. *Hint:* put a prior $\mu \sim \mathcal{N}(0, \lambda\mathbb{I}_n)$ and let $\lambda \to \infty$.

(d) Compute $\int_0^\infty \mathbb{E}R_\tau(\hat{s}_\tau)\,\mathrm{d}\tau$ for the plug-in estimator and compare it with the minimax rate $\widetilde{\Theta}(N^{-2/(n+4)})$ for estimating a Lipschitz score of a subgaussian density on $\mathbb{R}^n$ (Wibisono–Wu–Yang, 2024).

Exercise 6.10 (Bias–variance choice of the KDE variance ($\star$))**.** Write φ_v for the density of $\mathcal{N}(0, v\mathbb{I}_n)$. Let q be a density on $\mathbb{R}^n$ with $\|q\|_\infty < \infty$, C^1 and with L-Lipschitz score $s = \nabla \log q$ satisfying $\int |s|q < \infty$. Let $X \sim q$, $Y = X + \sqrt{v}\,Z$, $q_v = \mathrm{Law}(Y)$, $s_v = \nabla \log q_v$, and, given $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} q$, let $\hat{q}_v(y) := \frac{1}{N}\sum_{i=1}^N \varphi_v(y - X_i)$ be the Gaussian kernel density estimator with variance v (bandwidth $\sqrt{v}$). The regularised score estimator is $\hat{s}(y) := \nabla\hat{q}_v(y)/\max\{\hat{q}_v(y), \varepsilon_0\}$ for a floor $\varepsilon_0 > 0$.

(a) Show that $s_v(y) = \mathbb{E}[s(X) \mid Y = y]$.

(b) Deduce the bias bound $\mathbb{E}|s_v(Y) - s(Y)|^2 \leq L^2 n v$.

(c) Show that for every y, $\mathbb{E}|\nabla\hat{q}_v(y) - \nabla q_v(y)|^2 \leq \frac{n\|q\|_\infty}{2(4\pi)^{n/2}} \cdot \frac{1}{Nv^{n/2+1}}$ and $\mathbb{E}|\hat{q}_v(y) - q_v(y)|^2 \leq \frac{\|q\|_\infty}{(4\pi)^{n/2}} \cdot \frac{1}{Nv^{n/2}}$.

(d) (Heuristic.) Assume that on the relevant region $q_v \geq c > 0$, the floor is inactive and $|s_v| \leq A$. Linearising $\nabla\hat{q}_v/\hat{q}_v$ around $\nabla q_v/q_v$, argue that the risk behaves like $L^2 n v + B/(Nv^{n/2+1})$ for $v \leq 1$ and some B depending on $n, c, A, \|q\|_\infty$, and find the optimal v and the resulting rate.

Exercise 6.11 (A bump lower bound ($\star$))**.** Fix $n \geq 1$ and $L > 0$. Assume that there is a subgaussian density q_0 on $\mathbb{R}^n$ whose score is $(L/2)$-Lipschitz and which is constant, $q_0 \equiv c_0 > 0$, on a neighbourhood of the cube $Q = [0,1]^n$ (e.g. $q_0 \propto e^{-f}$ with f smooth, $f = 0$ near Q and $f(x) = a|x|^2$ for large $|x|$). Let $\psi \in C_c^\infty(\mathbb{R}^n)$ be supported in the open ball $B(0, 1/2)$, with $\int\psi = 0$, $\|\psi\|_\infty \leq 1$ and $\psi \not\equiv 0$. For an integer $M \geq 1$ let $r = 1/M$, let $x_1, \dots, x_m$ ($m = M^n$) be the centres of the cubes Q_j of side r partitioning Q, and set $\psi_j(x) = \psi((x - x_j)/r)$. For $\theta \in \{-1, 1\}^m$ and $\epsilon \in (0, 1/2]$ let

$$q_\theta := q_0\Big(1 + \epsilon\sum_{j=1}^m \theta_j\psi_j\Big).$$

(a) Show that q_θ is a probability density, equal to q_0 outside Q and $\geq c_0/2$ on Q, and that its score is Lipschitz with constant at most $L/2 + C_\psi\epsilon/r^2$, where $C_\psi := 2\|\nabla^2\psi\|_\infty + 2\|\nabla\psi\|_\infty^2$.

(b) Show that if θ, θ' differ only in coordinate j, then $\mathrm{KL}(q_\theta^{\otimes N} \,\|\, q_{\theta'}^{\otimes N}) \leq 8c_0\|\psi\|_2^2\, N\epsilon^2 r^n$.

(c) Show that, for such θ, θ' and any vector field g, $\int_{Q_j}|g - s_\theta|^2 q_\theta + \int_{Q_j}|g - s_{\theta'}|^2 q_{\theta'} \geq c_0\|\nabla\psi\|_2^2\,\epsilon^2 r^{n-2}$, where $s_\theta = \nabla\log q_\theta$.

(d) (Assouad's argument.) Using $\mathbb{E}_P f + \mathbb{E}_{P'} g \geq a(1 - \mathrm{TV}(P, P'))$ for measurable $f, g \geq 0$ with $f + g \geq a$, show that every estimator $\hat{s}$ satisfies $\max_\theta \mathbb{E}_\theta \int|\hat{s} - s_\theta|^2 q_\theta \geq \frac{m}{2} c_0\|\nabla\psi\|_2^2\epsilon^2 r^{n-2}(1 - \max \mathrm{TV})$, the maximum being over neighbouring pairs.

(e) Choose $\epsilon = Lr^2/(2C_\psi)$ and $M \asymp N^{1/(n+4)}$. Conclude that the minimax risk

$$\inf_{\hat{s}} \sup_{q} \mathbb{E} \int |\hat{s} - \nabla \log q|^2 q$$

over subgaussian densities q with L-Lipschitz score, the infimum being over estimators built from $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} q$, is at least $c\, N^{-2/(n+4)}$ for N large, with $c > 0$ depending on n, L, ψ and c_0.

Setting for Exercises 6.12–6.14. We write $m_\tau(x) = \mathbb{E}[\bar{X}_0 \mid \bar{X}_\tau = x]$ for the denoiser, so that Tweedie's formula reads $s_\tau(x) = (e^{-\tau} m_\tau(x) - x)/\sigma_\tau^2$, and $a(\tau) = \mathbb{E}|\bar{X}_0 - m_\tau(\bar{X}_\tau)|^2$. For $0 < \tau < \tau'$ the *DDPM score approximation* freezes the denoiser instead of the score,

$$s_\tau(x \mid \tau', x') = \frac{e^{-\tau} m_{\tau'}(x') - x}{\sigma_\tau^2},$$

whereas the *exponential integrator* approximates $s_\tau(x)$ by $s_{\tau'}(x')$. On a grid $0 = t_0 < \cdots < t_K = T - \delta$ we write $\tau_k = T - t_k$. With a (learned or exact) score $\hat{s}$, one step of the *DDPM sampler* reads

$$\hat{Y}_{t_{k+1}} = e^{h_k} \hat{Y}_{t_k} + e^{h_k} \sigma_{h_k}^2 \, \hat{s}_{\tau_k}(\hat{Y}_{t_k}) + \sigma_{h_k} \frac{\sigma_{\tau_{k+1}}}{\sigma_{\tau_k}} \xi_k, \qquad \xi_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbb{I}_n).$$

Exercise 6.12 (DDPM is an exact one-step posterior ($\star$))**.** Let $0 < \tau < \tau'$ and $h = \tau' - \tau$.

(a) Show that, conditionally on $\bar{X}_0 = x_0$ and $\bar{X}_{\tau'} = y$,

$$\bar{X}_\tau \sim \mathcal{N}\Big(\frac{e^{-h}\sigma_\tau^2}{\sigma_{\tau'}^2}\, y + \frac{e^{-\tau}\sigma_h^2}{\sigma_{\tau'}^2}\, x_0, \quad \frac{\sigma_\tau^2 \sigma_h^2}{\sigma_{\tau'}^2} \mathbb{I}_n \Big).$$

(b) Show that one DDPM step from level τ' to level τ with the exact score, started at y, samples exactly the law in (a) with x_0 replaced by the denoiser $m_{\tau'}(y) = e^{\tau'}(y + \sigma_{\tau'}^2 s_{\tau'}(y))$.

(c) Show that the DDPM step has the exact posterior mean $\mathbb{E}[\bar{X}_\tau \mid \bar{X}_{\tau'} = y]$, and that

$$\mathrm{Cov}(\bar{X}_\tau \mid \bar{X}_{\tau'} = y) = \frac{\sigma_\tau^2 \sigma_h^2}{\sigma_{\tau'}^2} \mathbb{I}_n + \Big(\frac{e^{-\tau}\sigma_h^2}{\sigma_{\tau'}^2} \Big)^2 \mathrm{Cov}(\bar{X}_0 \mid \bar{X}_{\tau'} = y).$$

So DDPM underestimates the posterior covariance, and is exact for all y if and only if q is a Dirac mass.

(d) Fix $y, m \in \mathbb{R}^n$ and let R solve, for $u \in [0, h]$,

$$\mathrm{d}R_u = \Big[R_u + 2\, \frac{e^{-(\tau'-u)} m - R_u}{\sigma_{\tau'-u}^2} \Big] \mathrm{d}u + \sqrt{2}\, \mathrm{d}B_u, \qquad R_0 = y.$$

Show that R_h has the law (a) with $x_0 = m$. Deduce that the DDPM update is the exact solution over one step of the reverse SDE in which the score $s_{T-t}(\cdot)$ is replaced by the corrected score $s_{T-t}(\cdot \mid \tau_k, \hat{Y}_{t_k})$ (with m_{τ_k} replaced by the learned denoiser $e^{\tau_k}(x + \sigma_{\tau_k}^2 \hat{s}_{\tau_k}(x))$ if the score is learned).

Exercise 6.13 (In a flat model the exponential integrator pays $n - d$ and DDPM pays d ($\star$))**.** Let $q = \mathcal{N}(0, \mathbb{I}_d) \otimes \delta_0^{\otimes(n-d)}$ on $\mathbb{R}^d \times \mathbb{R}^{n-d}$, and write $x = (x_\parallel, x_\perp)$.

(a) Show that $q_\tau = \mathcal{N}(0, \mathbb{I}_d) \otimes \mathcal{N}(0, \sigma_\tau^2 \mathbb{I}_{n-d})$, $s_\tau(x) = (-x_\parallel, -x_\perp/\sigma_\tau^2)$, $m_\tau(x) = (e^{-\tau} x_\parallel, 0)$ and $a(\tau) = d\sigma_\tau^2$.

(b) Let $0 < \tau < \tau'$ and $\Delta = \tau' - \tau$. Show that the local error of the exponential integrator is

$$\mathcal{E}_{\mathrm{EI}}(\tau, \tau') := \mathbb{E}|s_\tau(\bar{X}_\tau) - s_{\tau'}(\bar{X}_{\tau'})|^2 = (n-d)\Big[\frac{1}{\sigma_\tau^2} - \frac{1}{\sigma_{\tau'}^2} + \frac{2(1-e^{-\Delta})}{\sigma_{\tau'}^2}\Big] + 2d\,(1-e^{-\Delta}).$$

(c) Show that the local error of DDPM is

$$\mathcal{E}_{\mathrm{DDPM}}(\tau, \tau') := \mathbb{E}|s_\tau(\bar{X}_\tau \mid \tau', \bar{X}_{\tau'}) - s_\tau(\bar{X}_\tau)|^2 = d\,\frac{e^{-2\tau}(\sigma_{\tau'}^2 - \sigma_\tau^2)}{\sigma_\tau^4},$$

and that the normal coordinates contribute nothing.

(d) Now consider a geometric step. Let $\kappa \in (0, 1/4]$, $\tau' \le 1$ and integrate over $\tau \in [\tau'/(1+\kappa), \tau']$. Show that there are absolute constants $c, C > 0$ with

$$\int_{\tau'/(1+\kappa)}^{\tau'} \mathcal{E}_{\mathrm{EI}}(\tau, \tau')\mathrm{d}\tau \ge c\,(n-d)\kappa^2, \qquad \int_{\tau'/(1+\kappa)}^{\tau'} \mathcal{E}_{\mathrm{DDPM}}(\tau, \tau')\mathrm{d}\tau \le C\,d\kappa^2.$$

Conclude that over N such steps the exponential integrator accumulates at least $c(n-d)\kappa^2 N$ and DDPM at most $Cd\kappa^2 N$; with $N \asymp \kappa^{-1}\log(1/\delta)$ steps down to level δ, this is $(n-d)\kappa\log(1/\delta)$ versus $d\kappa\log(1/\delta)$.

Exercise 6.14 (Fixed-grid errors add across coordinates ($\star$))**.** Fix a grid $0 = t_0 < \cdots < t_K = T - \delta$ and consider the DDPM sampler with the *exact* score ($\hat{s} = s$) and *exact* initialisation ($\hat{Y}_0 \sim q_T$). Let π be a compactly supported probability on $\mathbb{R}$, and suppose that in dimension one with target π this sampler outputs a law ρ_π with $\mathrm{KL}(\pi_\delta \| \rho_\pi) = \bar{\varepsilon}$, where π_δ is the law of the OU process at level δ started from π. Let $1 \le d \le n$ and $q = \pi^{\otimes d} \otimes \delta_0^{\otimes(n-d)}$ on $\mathbb{R}^n$.

(a) Show that $q_\tau = \pi_\tau^{\otimes d} \otimes \mathcal{N}(0, \sigma_\tau^2)^{\otimes(n-d)}$, that s_τ acts coordinatewise, and that the coordinates of $\hat{Y}_{t_k}$ are independent for every k.

(b) Show that for a coordinate with target δ_0 the sampler is exact: $\mathrm{Law}(\hat{Y}_{t_k}^i) = \mathcal{N}(0, \sigma_{\tau_k}^2)$ for all k.

(c) Deduce that $\mathrm{KL}(q_\delta \| \,\mathrm{Law}(\hat{Y}_{t_K})) = d\,\bar{\varepsilon}$. Conclude that, on a fixed grid, no bound for this sampler of the form $o(d)$ (uniformly over such data, with constants independent of d) can hold. Why does this not show that the number of steps needed to reach a given accuracy must grow linearly in d?